

YIXUAN ZHANG | Asher

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EDUCATION

University of Illinois Urbana-Champaign

M.S. in Computer Science

Champaign, IL
Aug 2026 – May 2028

University of Nottingham

B.Sc. (Hons) Computer Science, First Class · GPA 3.88/4.00

Nottingham, UK
Sep 2022 – Jun 2026

Honors: Outstanding Graduate (Top 5%); Dean's Scholarship (Top 10%); Zhejiang Provincial Government Scholarship (Top 5%); MCM Honorable Mention.

Coursework: Computer Vision (ranked 1st), Computer Security (ranked 1st), Operating Systems, Algorithms, Databases, Artificial Intelligence.

TECHNICAL SKILLS

Languages: Python, C++, Java, SQL, TypeScript, Swift

ML / DL: PyTorch, Hugging Face Transformers, scikit-learn, OpenCV, computer vision, NLP, LoRA, RAG, ONNX, TensorRT, vLLM

Systems: FastAPI, Docker, Kubernetes, MySQL, MLflow, GitHub Actions, AWS, GCP, Linux

EXPERIENCE

Ningbo AST Machinery Technology

Software Engineer Intern | github.com/asher0913/multi-material-slicer

Ningbo, China
May 2026 – Aug 2026

- Shipped a cross-platform multi-material 3D-printing slicer in C++17, Qt, and OpenGL (STEP/STL assemblies, per-part materials, layer masks, G-code) with CMake builds and an Xvfb end-to-end self-test over the full import-to-G-code path.
- Designed a 51-table MySQL CRM (107 foreign keys, 70 constraints) and 10 Vue/FastAPI modules; built an LLM pipeline that produced 345 faculty profiles and 1,127 deduplicated grant records.

Zhiyang Innovation Technology

Machine Learning Engineer Intern

Shanghai, China
May 2025 – Aug 2025

- Cut Jetson Orin Nano inference latency from 220 ms to 95 ms (2.3×) and model size from 87 MB to 14 MB (6.2×) at under 0.5 mAP loss by exporting to ONNX and deploying INT8 TensorRT FP16 behind FastAPI and Docker.
- Raised mAP@0.5 from 0.74 to 0.79 and recall 6.2 points while cutting false positives 12% on a 7:1 imbalanced defect set by benchmarking YOLOv8s, YOLOv11, Faster R-CNN, and EfficientNet-B3 on 2×A100 with focal loss and class weighting.
- Built the training pipeline for 120K+ utility-equipment images from three sources; automated label QA and OpenCV augmentation cut label noise 18% and data prep from 3 days to under 1, adding 5.6 mAP@0.5.

Beijing Corefire Technology

Software Engineer Intern

Beijing, China
May 2024 – Jul 2024

- Built merchant onboarding, query, and reporting services for UMFintech's acquiring platform in Spring Cloud and MySQL, covering hierarchical providers, POS/QR transactions, terminal lifecycle, and profit-sharing summaries.

PROJECTS

DualPath-CEM | github.com/asher0913/DualPathCEM · PyTorch, privacy ML

Nov 2025 – Jun 2026

- Reached 81.96% mean top-1 on a 44,139-image FaceScrub split (526 identities, 5 seeds), 1.63 points above the published Noise_ARL+CEM reference, while raising decoder and GAN reconstruction MSE 75.8% and 74.9%.
- Designed dual-path split inference: a Gaussian-protected 16×16×16 SlotCEM tensor plus a 256-D MobileNetV3 token, fused by calibrated logits; SlotCEM projects 4,096-D smashed features into 8 slots of 64-D across a 64-entry per-class memory.

Manuscript in preparation: Y. Zhang and J. Ren, "DDP-CEM: Dual-Path Privacy Protection for Collaborative Face Inference," *Pattern Recognition*, 2026.

SkillRouter | github.com/asher0913/skill-router · Python, LLM agents

Sep 2026

- Cut per-query agent context from 14,108 to 2,052 tokens (−85.5%) on a 60-skill catalogue while holding Top-1 routing accuracy at 0.984 and Top-3 at 1.000 by tiering each skill and expanding only the selected ones; 26 tests in GitHub Actions.

InferenceLab | github.com/asher0913/inference-lab · Python, vLLM, MLOps

Jul 2026 – Sep 2026

- Built an async dynamic batcher and TTL/LRU semantic cache behind a swappable vLLM/SGLang/Ollama backend; an 80-request, 16-concurrency workload collapsed into 2 backend batches at 80% cache-hit rate and ~79 ms P95.

SecureRAG | github.com/asher0913/secure-rag · FastAPI, RAG, ACL

Feb 2025 – Apr 2025

- Implemented multi-tenant hybrid retrieval that applies tenant/ACL filters before ranking and repeats live permission checks before context assembly, failing closed on revocation; added Recall@K, MRR, and authorization-leakage regression tests.